

Model Failure Challenge Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	Time	25-35 min
Difficulty	Medium	ID	model-failure-challenge

Big idea: Models simplify reality productively.

Student-Facing Prompt

Inspect Model Failure Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Model Failure Challenge Stimulus A	Student-facing stimulus 1: A simple weather, disease, or traffic model that works until one assumption changes. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Model Failure Challenge Stimulus B	Student-facing stimulus 2: An anomaly card that forces students to revise a model. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Model Failure Challenge Reveal / Twist	Student-facing stimulus 3: A model limits table: useful for, weak when, assumptions, and missing variables. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Models simplify reality productively.

Actual Lesson Questions

#	Question for students
1	After inspecting Model Failure Challenge Stimulus A, what is your first knowledge claim?
2	Which exact detail from Model Failure Challenge Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Are models useful because they are false?

Ready-to-Use Examples

- A simple weather, disease, or traffic model that works until one assumption changes.
- An anomaly card that forces students to revise a model.
- A model limits table: useful for, weak when, assumptions, and missing variables.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Groups identify variables missing from the model: prior knowledge, sleep, stress, revision quality, teaching, test difficulty.
2. They draw a revised model with extra variables.
3. They decide whether the original model is useless, useful or dangerous.
4. Discuss when a model should be rejected.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Are models useful because they are false?
- How much simplification is acceptable?
- What is the difference between a model, a law and an explanation?

Knowledge Questions

- Are models useful because they are false?
- How much simplification is acceptable?
- What is the difference between a model, a law and an explanation?

Transfer Example

For example, run this activity with a simple weather, disease, or traffic model that works until one assumption changes. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...